

Practical Lab File for Agentic AI

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EXPERIMENT 2

Aim: Five Levels of Text Splitting

1. Introduction

This notebook explains **five different levels of text splitting techniques** used in **Natural Language Processing (NLP)**.

Text splitting is a crucial preprocessing step, especially in tasks like:

- Text analysis
- Information retrieval
- Chatbots
- Large Language Models (LLMs)
- Document indexing

The notebook gradually moves from **basic splitting** to **advanced intelligent splitting**.

2. Objective

The objectives of this notebook are:

- To understand why text splitting is important
- To explore multiple strategies for splitting text
- To compare different levels of text segmentation
- To apply text splitting techniques for NLP-based applications

3. Technologies & Libraries Used

- **Python**
- **LangChain (Text Splitters)**
- **NLP concepts**
- **Recursive algorithms**

4. Why Text Splitting is Required?

Large documents cannot be processed directly by:

- Language models (token limits)
- Search engines
- Vector databases

Therefore, text must be split into **smaller meaningful chunks** without losing context.

5. Level 1: Character-Based Text Splitting

Description:

This is the **simplest form** of text splitting.

Method:

- Splits text purely based on **number of characters**
- Does not consider sentence or paragraph meaning

Use Case:

- Quick chunking
- When structure does not matter

Limitation:

- ✗ May break sentences in the middle
- ✗ Context loss

6. Level 2: Recursive Character Text Splitting

Description:

An improved version of character-based splitting.

Method:

- Tries splitting text using separators in this order:
 1. Paragraphs
 2. Sentences
 3. Words
 4. Characters

Advantage:

- ✓ Preserves sentence structure
- ✓ More meaningful chunks

Common Use:

- LLM document preprocessing
- Knowledge base creation

7. Level 3: Token-Based Text Splitting

Description:

Splits text based on **tokens instead of characters**.

Method:

- Uses tokenizer logic (similar to GPT models)
- Ensures token limits are respected

Importance:

- LLMs work on tokens, not characters
- Prevents token overflow errors

Use Case:

- ChatGPT-based applications
- Embedding generation

8. Level 4: Document-Specific Text Splitting

Description:

Splitting strategy depends on the **document type**.

Examples:

- PDF → split by pages
- Code → split by functions/classes
- HTML → split by tags
- Markdown → split by headings

Advantage:

- ✓ Maintains semantic structure
- ✓ Highly accurate chunking

9. Level 5: Semantic Text Splitting

Description:

The **most advanced level** of text splitting.

Method:

- Splits text based on **meaning and context**
- Uses embeddings and similarity scores

Key Features:

- Groups related sentences together
- Avoids breaking logical ideas

Use Case:

- RAG (Retrieval-Augmented Generation)
- AI assistants
- Search engines

10. Comparison of All 5 Levels

Level	Technique	Accuracy	Complexity
1	Character-based	Low	Very Low
2	Recursive Character	Medium	Low
3	Token-based	High	Medium
4	Document-based	High	Medium
5	Semantic splitting	Very High	High

11. Applications

- Chatbots
- AI search engines
- Document summarization
- Question-answering systems
- Large language model pipelines

12. Conclusion

This notebook demonstrates that:

- Text splitting is **essential for NLP**
- No single splitting technique fits all use cases
- Advanced applications require **semantic-aware splitting**
- Choosing the right level improves accuracy and performance

13. Future Scope

- Combine semantic splitting with vector databases
- Optimize chunk sizes dynamically
- Apply hybrid splitting strategies
- Use AI-based split evaluation